

The impact of state financial incentives on market deployment of solar technology

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ABSTRACT

Many states have adopted financial incentives to encourage market deployment of solar energy technology. This paper employs a cross-sectional time-series approach to evaluate the extent to which state solar financial incentives systematically encouraged market deployment of solar photovoltaic (PV) technology from 1997 to 2009. The results demonstrate that states offering cash incentives such as rebates and grants experienced more extensive and rapid deployment of grid-tied PV technology than states without cash incentives over the study period. The analysis also finds that the presence of state renewable energy portfolio standards and specific solar carve-out provisions within them heavily influenced the market deployment of grid-tied solar PV technology through 2009.

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1. Introduction

Concern about the nation's energy supply and infrastructure is high. Families and businesses are feeling the pinch of higher energy costs, and reverberations have been felt throughout the economy. At the same time, climate researchers assert that the country must reduce its production of greenhouse gases or suffer potentially catastrophic effects from a changing climate. Many now assert that a shift towards clean energy is imperative to relieve energy burdens and achieve sustainability goals.

Solar technology could play an important role in greening the nation's energy portfolio. Decades of research, development, and demonstration (RD&D) have yielded viable solar technologies for on-site energy use and large-scale energy production. Yet the potential energy gains from solar technology remain elusive. Solar technology provides just 0.1% of the nation's current energy (U.S. Energy Information Administration [EIA], 2010, 2011). The initial cost of solar technology remains a persistent barrier to its widespread adoption.

Public policy has endeavored to address the cost barrier by investing in RD&D, subsidizing the purchase and manufacturing of solar technology, and adopting supportive regulatory policy. Much of the policy support for solar technology has come from states, showing substantial policy leadership and innovation as

they design and implement solar incentive programs (Sarzynski, 2009). The nation's large geographic area makes a one-size-fits-all approach to energy production and distribution inefficient. The availability and cost structure of other sources of energy varies regionally, making solar a more attractive option in some places than others. Climatic conditions also vary in ways that make solar more technically feasible in some parts of the country than others. More importantly, states hold the key to adoption of solar energy technology because they have primary regulatory authority over utilities and many policy areas relevant to energy investment and production, including building codes, land uses, consumer protection, and taxes (Friedlander and Sawyer, 1983).

Here, we are interested in the extent to which state financial incentives have spurred consumer adoption of solar technology. The research to date has found a positive impact of the presence of specific solar financial incentives (usually tax credits) on deployment of solar technology, using a variety of research methods and data sources. What is less well understood is how the entire suite of financial incentives (tax credits, rebates, etc.) may encourage solar energy use. In addition, there has been little attempt to distinguish effects of federal incentives from state or local incentives, or to distinguish the effect of tax credits from other programs. Likewise, there has been little recent analysis, with the bulk of studies dating to the 1970s and 1980s. Now that states have adopted a larger variety of programs, the opportunity arises to evaluate the impact of different types of incentives and differential implementation on solar energy use. In addition, the fiscal challenges facing most states today make incentive assessment and evaluation all the more important.

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This paper adds to the small body of research on comparative effectiveness of solar incentives by conducting a rigorous quantitative impact evaluation of state-level financial incentives on market deployment of grid-tied photovoltaics (PV) through 2009. We find that states with cash incentives such as rebates and grants experienced more extensive deployment of PV technology than states without cash incentives. The analysis also finds that the presence of state renewable energy portfolio standards and specific solar carve-out provisions within them heavily influenced the market deployment of solar technology.

1.1. Previous literature

Consumer choice theory provides a framework for understanding the adoption of solar technology (Lazzari, 1983). Solar technology is a durable good that substitutes for conventional energy technology typically powered by electricity or fossil fuels. Solar technology has a higher capital cost than conventional energy technologies but requires no fuel inputs and thus can deliver annual energy savings over conventional technology. Whether solar technology is cost-competitive with conventional technologies depends generally on the upfront cost of the solar technology, the expected savings of conventional energy (which depends on the type of conventional energy employed), and availability of any public subsidies for renewable technologies. Feiveson and Rabl (1982) argued that incentives to promote alternative energy are necessary to spur consumer adoption due to market failures in the conventional energy and utility markets. Direct subsidies to conventional fuels, fuel expensing for tax purposes, government control of natural gas prices, average rather than marginal cost pricing by utilities, and environmental externalities artificially force alternative energies like solar to be more expensive relative to other conventional energy sources.

Various economic studies have been performed to assess the sensitivity of the consumer's cost-effectiveness calculation for solar technologies to changes in energy alternatives, prices, incentives, or solar resources (Bezdek et al., 1979; Bezdek and Sparrow, 1981; Fry, 1986; Kastler, 1983; Proctor and Tyner, 1984; Ruegg, 1976). The approach suffers from one major drawback: it does not evaluate the actual behavior of consumers when faced with different energy conditions and incentives (Lazzari, 1983). Consumers may not necessarily act rationally even when energy-saving technology could generate substantial net benefits (Quigley, 1991). The few scholars that have examined the impact of incentives on observed consumer behavior found that the availability of federal or state income tax incentives increased the likelihood of installing solar technologies (Durham et al., 1988; Hassett and Metcalf, 1995).

Since these early studies, the market for solar technology has changed considerably and state-level policy activity has increased. This paper contributes a much-needed comprehensive evaluation of the effectiveness of solar incentives, using up-to-date data and rigorous econometric techniques. Several recent econometric studies have examined the impact of state energy policies on the penetration of renewable technologies (Carley, 2009; Menz and Vachon, 2006; Yin and Powers, 2010). These studies mostly focus on wind energy, which constitutes more than 90% of non-hydropower renewable energy. Further, these studies focus primarily on renewable energy mandates. Our study adds to the literature by focusing specifically on solar technology and on state-level financial incentives, which have yet to be subject to extensive study in recent years.

1.2. Methods and data

1.2.1. Empirical framework

The purpose of this paper is to evaluate the systematic and historical impact of state-level financial incentives on the market

deployment of solar technology. We employ a time-series cross-sectional (TSCS) fixed-effects model that accounts for unobserved state-level and time variation in solar technology deployment, with the following form:

$$PVINS_{it} = \alpha_0 + \beta I_{it} + \gamma C_{it} + S_i + T_t + \varepsilon_{it}$$

where for state i and year t , $PVINS_{it}$ represents the annual amount of solar technology installed, I_{it} represents a vector of state financial incentives available, C_{it} represents a vector of control variables including other state energy policies, electricity prices, income, and population, S_i represents a vector of state dummy variables that are time-invariant, and T_t represents a vector of time dummy variables. The error term, ε_{it} , is comprised of a time-invariant state fixed effect and an independent and identically distributed random state-year term, as follows:

$$\varepsilon_{it} = \mu_i + \eta_{it}$$

The vector of coefficients β provides a quantitative measurement of the extent to which a type of financial incentive systematically encouraged adoption of solar technology during the study period. Such an approach captures the aggregate behavior of consumers within a state. Thus, the model infers the impact of different factors on aggregate behavior but is not sufficient to predict behavior by an individual consumer. The model also infers systematic and historical effects across all states and is not sufficient to assess the effectiveness of any particular state's financial incentive program. For this reason, this research supplements case study research on individual state programs.

1.3. Dependent variable

This analysis examines the annual amount of grid-tied photovoltaic capacity installed in a state (in kilowatts, kW), from 1997 to 2009 ($PVINS$). The U.S. Department of Energy's Energy Information Administration (EIA) has not historically tracked PV shipments by destination state as it has with solar thermal technology, although it began to do so in 2007. Data on annual grid-tied PV installations were instead obtained from the Interstate Renewable Energy Council (Sherwood, 2011). IREC tracked only grid-tied PV installations, meaning those installations that were connected to the electricity grid, and thus roughly 10–15% of all PV installations were not included in this dataset (Sherwood, 2011).

At the national level, grid-tied PV installations increased exponentially from a negligible amount in 1997 to over 430,000 kW in 2009 (Fig. 1). PV technology has been deployed to generate on-site electricity for residences, commercial and industrial establishments, and recently for utility-scale installations. The majority of PV installations have been for small residential systems, although much of the PV installed capacity (in kW) has been from larger commercial and utility-scale systems (Sherwood, 2011).

Other solar technologies such as for residential heating and cooling have been concentrated in states with strong sunlight, with Florida, California, Hawaii, and Arizona capturing more than 80% of the installations nationwide through 2009. Grid-tied PV capacity has been even more concentrated, with California comprising more than 60% of the U.S. PV market through 2009. Nevertheless, PV capacity has expanded recently to other states with strong sunlight, such as Hawaii and Nevada, as well as to less expected places like Colorado, New Jersey, Oregon, and the New England states (Table 1). The differential deployment across states suggests a role for public policy in facilitating solar PV deployment.

The evolution of PV installations from 1997 to 2009 also varied across the states. Some states experienced consistent growth in PV since the mid-2000s, while other states have seen very slow growth in PV across the entire study period. A noticeable increase

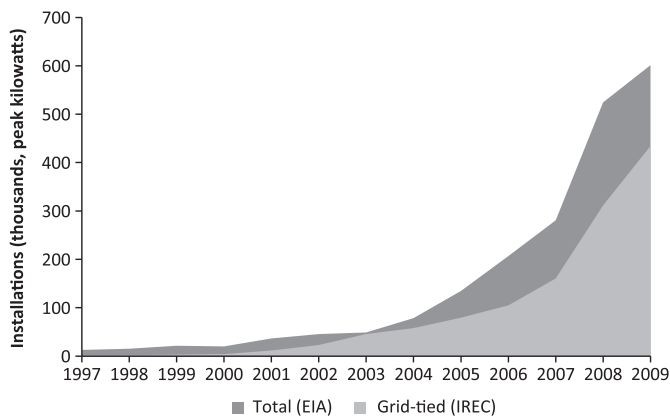


Fig. 1. Annual U.S. PV installations (1997–2009).

Source: Sherwood, 2011 (grid-tied); U.S. Energy Information Administration [EIA], 2011 (total).

Table 1

Average cumulative per capita PV installations (1997–2009).

Source: Sherwood, 2011 and authors' calculation.

Cumulative per capita PV installations (Wdc/person) (1997–2009)	State
> 20	CA, HI
10 < ... < 20	CO, NJ, NV
2 < ... < 10	AZ, CT, DE, FL, MA, OR, VT
0 < ... < 1	AL, AR, DC, GA, IA, ID, IL, IN, KY, LA, MD, ME, MI, MN, MO, MS, MT, NC, NH, NM, NY, OH, OK, PA, RI, SC, TN, TX, UT, VA, WA, WI, WY
0 ^a	AK, KS, ND, NE, SD, WV

^a 0 or missing data.

in recent installations was common to most states, likely reflecting higher energy prices and increased availability of state and federal solar incentives.

Recent deployment trends reflect rapidly falling costs for solar PV systems. PV module prices have fallen in the U.S. from approximately \$9.40/W in 1990 to \$2.80/W in 2009 (U.S. Energy Information Administration [EIA], 2011). Installed PV systems in the U.S. cost an average of \$6.2/W in 2010 (Wiser et al., 2011). Despite falling prices, solar-generated electricity remains capital intensive for the purchaser and more expensive than conventional electricity generation, suggesting the continued need for financial incentives to spur market deployment.

1.4. State-level financial incentives

This study focuses primarily on the presence of state-funded financial incentives for solar technology within a state in a given year. Higher annual installations are expected in states where incentives effectively persuaded consumers to adopt solar technology. Information on the presence of state-level financial incentives was collected from many sources. Currently-offered incentives are tracked in the *Database of State Incentives for Renewable Energy* (DSIRE). While DSIRE tracks the initial adoption dates for incentives, it does not include older incentives that may have expired. Various reports and papers were consulted to fill in gaps and verify adoption dates. While every effort was made to determine the presence of state-level incentives for each year analyzed here, it is possible that the presence of some incentive programs have been overlooked. Major incentive programs,

especially offered through the tax system, are believed to be tracked properly. Any effect of omitted incentives should be small and should not significantly alter the results.

Four types of financial incentives for consumer adoption of solar technology are distinguished here: income tax incentives; cash incentives; sales tax incentives; and property tax incentives. Income tax and cash incentives provide consumers with a subsidy to install solar technology, typically claimed within one year of purchase although sometimes are tied to the amount of electricity generated over a 10- or 15-year period. Sales tax incentives provide partial or total sales tax relief to purchasers of solar technology, typically claimed at time of purchase. Property tax incentives provide partial or total property tax relief to owners of real properties where solar technology is utilized, preventing an increase in property value assessments after installation. We expect that incentives claimed at the time of purchase or within a short period of time after purchase would factor more prominently into a consumer's decision-making process than incentives that could be claimed in later years, due to the high upfront cost of solar technology, inflationary effects that devalue future incentives, and uncertainty about availability of future incentives.

Few states offer the same financial incentive designs. Instead, the incentives vary widely by function, type, method used to calculate incentive value, maximum incentive value, eligible technology, or sectors that may claim the incentive (Sarzynski, 2009). A few comparisons are instructive. Arizona offers a personal income tax credit for households at 25% of the installed cost, with a maximum credit of \$1000. By contrast, Hawaii offers a personal income tax credit of 35% of the installed cost, with a maximum credit of \$5000 for PV. Thus, residents in Hawaii get a larger magnitude income tax incentive and a higher maximum credit, together providing a larger total incentive than for the same installation in Arizona. The substantially higher electricity prices in Hawaii work with the financial incentive to make solar technology much more cost-effective for the consumer in Hawaii than in Arizona (Sarzynski, 2010). Alternatively, states like Connecticut and New Jersey have employed cash rebates to support PV, and in some cases have spent upwards of \$20,000 per installation to make residential PV a cost-effective investment for consumers (Sarzynski, 2010).

Additional financial incentives may be offered by states to support manufacturing or RD&D activities. This analysis considers only the financial incentives offered to end-users of the technology. The analysis also considers only incentives that are paid by the state, and excludes incentives offered by individual utilities, localities, or state-wide non-profit organizations. Nevertheless, the effect of other incentives may inflate the apparent impact of the state-funded incentive in situations where state-funded incentives are also offered. Future analysis would need to map the availability of other incentives in order to assess the extent to which each state's consumers are capable of obtaining a non-state incentive.

Each state may have multiple incentives of any particular type, such as residential and business income tax credits, or an up-front cash grant for households and an annual payment for businesses based on the amount of solar electricity produced. In addition, each incentive may vary widely in its size and scope. Here, a type of financial incentive was represented as present if any incentives of the type were present. This representation hides some richness in policy design. This choice was a reasonable first-order approximation for studying the impact of policy on the deployment of solar technology, and was necessary because comprehensive information on state incentive policy designs offered each year back to the mid-1990s do not exist. Nevertheless, this simplification means that our analysis may compound effects of policy design with policy use.

Originally, state-funded solar financial incentives were enacted in response to the 1973 oil crisis. The crisis led to a sharp fall in crude oil supply from the Middle East, and consequently a spike in the global price of petroleum, peaking in 1979/1980 and remaining high until 1986. In 1986, as the crude supply came back online, it was met by lower demand due to years of crisis and the price of oil plummeted. President Reagan also dismantled various Carter-era energy policies and subsidies, including federal income tax incentives for solar technology, RD&D support of solar technology, and grants to state energy offices (Frankel, 1986). As a result of these and other factors, the number of states offering incentives for solar technology dropped off dramatically in the early 1980s. It was not until the mid-1990s – when higher energy prices and environmental legislation again led to increased interest in solar energy – that the number of states with solar financial incentives began to increase (Fig. 2) (Young and Sarzynski, 2009).

Early solar incentives were designed for low-temperature solar thermal collectors that were typically used for heating or cooling in residences. Over time, the incentives expanded to include emerging technologies that derived energy from sunlight, such as PV and higher-temperature solar thermal collectors that could be used on commercial, industrial, or utility-scale installations. Some states regularly adjusted the size or design of their incentives over time to respond to the changing technologies and their differential price structures. This analysis presumes that all state solar incentives were available for grid-tied PV technology even if their size was targeted towards cheaper solar thermal systems.

In 1997, 31 states offered some form of incentive for solar technology. Property and sales tax incentives were most prevalent, with only a few states offering cash or income tax incentives. By 2009, 46 states offered at least one type of solar financial incentive. Seven states offered all four types of incentives: Massachusetts, Maryland, New York, Ohio, Oregon, Rhode Island, and Vermont. Most states offered one or two different incentive types. Four states offered no financial incentives for consumer adoption of solar technology in 2009: Alabama, Arkansas, Missouri, and Mississippi. Property tax incentives remained the most widely utilized in 2009, being available in 29 states. Cash incentives were available in 28 states, income tax incentives were available in 26 states, and sales tax incentives were available in 25 states. Note that five states do not collect state sales taxes (Alaska, Delaware, Montana, Oregon, and New Hampshire) and were denoted as offering a sales tax incentive for the entire study period.

1.5. Other policy variables

One competing explanation for recent solar deployment is the presence of the federal investment tax credit for solar technology.

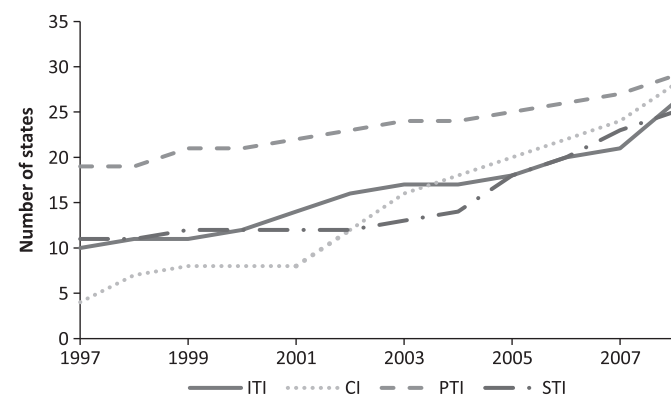


Fig. 2. Evolution of state-level solar financial incentives by incentive type (1997–2009). Notes: ITI=income tax incentive; CI=cash incentive; PTI=state-level property-tax incentive; STI=sales tax incentive.

A federal residential income tax credit was originally adopted in 1978 to reduce household energy use in the wake of the oil price shocks of the 1970s. As real energy prices declined in the early 1980s, many federal energy policies came under increasing scrutiny and the residential credit was discontinued at the end of 1985. The federal residential credits were reinstated in 2005 during a period of energy price escalation and concern over dependence on foreign oil-conditions similar to those in 1978. The federal income tax credit was expanded to non-residential users in 2006. Since the federal income tax credit does not vary between states, it was not included in our analysis, although its effects are captured primarily in the time fixed-effects dummies for the years 2006–2009.

Another competing explanation for state-level variation in consumer adoption of renewable technology is the adoption of renewable portfolio standards (RPS). RPS's generally mandate a certain share of the total electricity sold in a state to come from renewable energy sources. Some state RPS's have a "set-aside" amount of the total that must come from solar technology (also known as a "carve-out"), or require utilities to obtain some of their renewable energy from distributed, customer-sited technologies, such as grid-connected PV. Thus, a dummy variable for the presence of a state-level RPS (RPS) was included in the model, coded as 1 for years in which an RPS was effective and 0 for years in which it was not effective. A separate dummy variable for the presence of a solar carve-out (CARVE) was included, coded as 1 for years in which the carve-out was effective and 0 for years in which it was not effective. Data on the adoption of state RPS policies were obtained from Wisner and Barbose (2008) and from DSIRE.

A third potential explanation for changes in market deployment of PV installations is the presence of state net-metering policies. Net-metering allows customers who generate their own electricity through distributed technology like solar PV to bank excess electricity on the grid using bi-directional meters. In return, the customer receives energy credits that can be used to offset part or all of the electricity purchased from the utility within a certain period of time. Net-metering policies are important because they can represent a direct reduction in the price the consumer pays for electricity purchased from the utility, without requiring any public financing. In some states, customers can sell renewable energy credits (RECs) to the utility company or other entities to help meet the state's RPS requirements, further improving the financial benefits the customer receives from installing a PV system. A dummy variable was used to indicate the presence of a state-wide net-metering policy (METER), coded as 1 for each year in which net-metering was available to consumers statewide and 0 for years it was not available. Data on the adoption of state net-metering policies were obtained from DSIRE. Note that some utilities may offer net-metering even without state-wide net-metering policies.

The adoption of other state-level energy policies tracks well with the adoption of state financial incentives for solar energy. In 1997, only 10 states offered state-wide net-metering and only 6 states had a state-wide renewable portfolio standard in place (Fig. 3). By 2009, 43 states had adopted net-metering policies and 34 states had adopted a renewable portfolio standard (RPS). Ten states had specific solar provisions as part of their RPS policies in 2009.

1.6. Control variables

Consumer choice theory provides several alternative explanations for market deployment of solar technology, including the price of solar technology, the price of conventional energy, and consumer income.

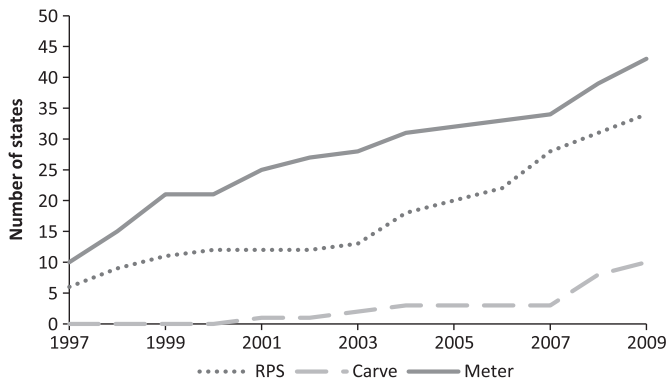


Fig. 3. Evolution of other state-level energy policies (1997–2009). Notes: RPS=renewable portfolio standard; Carve=solar carve-out in RPS; Meter=net-metering.

The falling price of solar technology will improve the potential return on investment for solar technology. Final installation costs for PV systems vary significantly across states depending on local labor and installation costs (Wiser et al., 2011). Unfortunately, historical PV module and installation costs are only available as national averages. For this reason, a PV price variable was not included in the model. The effect of changes in module and installation costs on the market deployment of PV will be captured primarily by the state and year fixed effects.

High or volatile conventional energy prices improve the potential return on investment for solar technology, as consumers look to substitute away from commodities with volatile prices. A variable was included in the model for the average price of electricity for the commercial sector (*ESCCD*), in dollars per million Btu. Commercial prices were used because the majority of domestically-produced PV capacity was shipped to commercial customers. Energy price data were obtained by state from the DOE's Consumption, Prices, and Expenditures estimates, which are published annually in the EIA's *State Energy Data System* (*SEDS*). The purchasing power of money varies substantially over the time period studied in this analysis. For this reason, energy price data were adjusted to 2008 dollars using the *Consumer Price Index for All Urban Consumers* from the Bureau for Labor Statistics. Note that the energy price data used here do not account for other pricing issues that may influence electricity savings and consumer decision-making, such as time-of-use pricing, tiered rate structures, or monthly minimums.

Wealthier consumers may be more likely to purchase and use solar technology regardless of incentives or other factors. No consistent information is collected across states and over time regarding the income of actual solar technology consumers. Instead, the model included a variable for the gross domestic product (GDP) per capita within a state in a given year (*PCGDP*), in thousands of chained 2005 U.S. dollars. The assumption is that states with higher per capita GDP may have more consumers that could afford to invest in solar technology, with its high upfront costs. Wealth also may serve as a proxy for propensity to adopt innovative technology. Previous studies have also found that solar income tax credits are more often claimed by middle- and upper-income filers than low-income filers (Lazzari, 1983; Quigley, 1991; Rodberg and Schachter, 1980). Annual state-level per capita GDP data were obtained from the *SEDS*.

The model includes a control variable for population of the state, on the premise that annual PV installations may increase with population size. Annual state-level population data (*TPOPP*), in thousands of residents including armed forces, were also obtained from the *SEDS*.

Electricity prices, per capita GDP, population, and the dependent variable exhibited non-normal distributions across states in

a given year, and were normalized using a natural logarithmic transformation. The resulting regression coefficients can be interpreted as elasticities, meaning the percent change in PV installations as a result of a 1% change in the electricity price, per capita GDP, or population.

The dependent variable was normalized by taking the natural log of $(1 + PVINS_{it})$ to retain observations for states that did not have any reported installations in some years. This adjustment is appropriate because the dependent variable can be equal to or greater than zero, and the natural log of zero is undefined. While the use of 1 as a constant is typical in this situation, the results may be sensitive to the choice of constant.

Finally, the model does not explicitly control for variation in solar insolation (i.e., the strength of sunlight) across states, because insolation does not vary over time. The effects of solar insolation are picked up by the state dummy variables.

1.7. Model specifications

Two model specifications were tested, as follows:

$$PVINS_{it} = \alpha_0 + \beta_1 ITI_{it} + \beta_2 CI_{it} + \beta_3 STI_{it} + \beta_4 PTI_{it} + \gamma C_{it} + S_i + T_t + \varepsilon_{it} \quad (1)$$

$$PVINS_{it} = \alpha_0 + \beta_1 YRITI_{it} + \beta_2 YRCI_{it} + \beta_3 YRSTI_{it} + \beta_4 YRPTI_{it} + \gamma C_{it} + S_i + T_t + \varepsilon_{it} \quad (2)$$

In the first specification, dummy variables were used to indicate the presence of four types of state financial incentives for solar technology: income tax incentives (*ITI*); cash incentives (*CI*); sales tax incentives (*STI*); and property tax incentives (*PTI*). By including separate variables for the different types of incentives, it is possible to empirically test for the impact of any one incentive type as well as to test the combined impact of multiple incentives (Gouchie et al., 2002). For each type of incentive, the dummy variable was coded as 1 for each year in which the incentive was available to consumers and 0 for years in which the incentive was not available.

The second specification tested the years since adoption for the four types of financial incentives (denoted as *YRITI*, *YRCI*, *YRSTI*, *YRPTI*). The second specification tested the hypothesis that the impact of a state-level incentive strengthens over time, as knowledge of the program spreads or as the incentive is broadened to cover more potential applicants (Howell-Moroney, 2007). In the second specification, the duration variables were coded as 1 in the first year in which the incentive became available, 2 for the second year available, and so on. The other three policy variables were also coded as duration variables, on a similar hypothesis that their effect strengthens over time (denoted as *YRRPS*, *YRCARVE*, *YRMETER*).

1.8. Model estimation

Given apparent heterogeneity across states, a series of Hausman tests were performed on Models (1) and (2) to determine if fixed effects (FE) were necessary. The Hausman tests overwhelmingly rejected the null hypothesis of no unit heterogeneity, confirming the need to control for unobserved differences between states that cannot be captured by the explanatory variables. For this reason, the models include state dummy variables as explanatory variables. These dummy variables allow the constant term to change for each state, thus avoiding any heterogeneity bias that would have occurred from imposing a common constant term for all states.

The nearly exponential increase in PV deployment over the period hints at the existence of unobserved time-specific influences on solar consumption that are common to all states. These

influences likely include falling solar PV system costs, changes in awareness of environmental issues, as well as recessions or presidential elections. As a result, time fixed effects via year dummy variables were included in both models.

Time-series cross-sectional models often suffer from additional data concerns, including serial correlation, panel heteroskedasticity, and contemporaneous correlation of the errors. A Wooldridge test for first-order autocorrelation in panel data (Wooldridge, 2002) could not reject the presence of serial-correlated errors in the FE models. Moreover, a modified Wald test for group-wise heteroskedasticity could not reject the presence of heteroskedastic errors in the FE models, and a Fries test could not reject the presence of contemporaneous correlation among the cross-sections. Consequently, Prais–Winston (PW) estimation with Panel Corrected Standard Errors (PCSE) was applied to both models. PCSE uses ordinary least squares coefficient estimates and residuals to compute standard errors that correct for panel heteroskedasticity (Beck and Katz, 1995). This approach has been widely applied to political science and public policy econometric analysis because of the prevalence of panel heteroskedasticity. Monte Carlo simulations described in Beck (2001) demonstrated that PCSE outperforms feasible generalized least squares (FGLS) in the presence of panel heteroskedasticity and contemporaneously correlated errors, and performed as well as FGLS in its absence.

1.9. Empirical results

Table 2 presents the results from the regression analysis. Models (1) and (2) present the effect of the four types of state financial incentives on grid-tied PV installations from 1997 to 2009 under two specifications. Model (1) tests for the average effect of having a financial incentive in any given year, while Model (2) tests for the incremental effect of offering an incentive for an additional year.

Our results indicate that states offering cash incentives, such as rebates or grants, had consistently stronger deployment of PV technology from 1997 to 2009 than states not offering cash incentives. On the other hand, states offering tax incentives such as income, sales, or property tax incentives did not experience

systematically stronger deployment of PV technology over the study period than states not offering tax incentives. Recall that the effect of cash incentives in the years 2006–2009 should be most appropriately attributed to the combined presence of state and federal incentives in these years, rather than simply to state cash incentives.

The impact of cash incentives on PV installations appears quite important to market deployment. Holding all other variables constant, the presence of a cash incentive in any given state was associated with a 248% higher amount of PV installed, on average, as compared to states not having a cash incentive (computed using Halvorsen and Palmquist, 1980 methodology). In the subsequent years following the adoption of a cash incentive, states with cash incentives installed an additional 23% more PV capacity per year, on average, as compared with states not offering cash incentives.

We suspect that the differential effects of the cash vs. tax incentives may relate to the effective size of the incentive and to the ease with which it can be claimed. Cash incentives – often connected to state RPS policies or to public benefit funds – tend to be large in value and may be sufficient to make PV investments financially sound, especially when combined with the federal investment tax credit (Sarzynski, 2010). Cash incentives also tend to be simple in design and easy to claim by the purchaser within a short period of time after purchase (i.e., 90 days).

In principle, income tax incentives should function as well as cash incentives, especially when combined with the federal income tax credit, given that they typically offset the upfront cost of purchasing the technology and can be easy to claim by tax filers. Yet, our results suggest that states with income tax incentives did not experience systematically more deployment of PV technology than states without income tax incentives. This result may be due to insufficient tax liability and to the limited size and scope of these tax incentives. Income tax incentives assume that the buyer has income tax liability that can be offset via the use of the income tax credit or deduction. This design neglects non-profit organizations or claimants with insufficient tax liability, and has been especially troublesome in recent years. Further, many state income tax incentives have not been of a sufficient size on their own to make PV a viable financial investment option for most consumers until very recently (Sarzynski, 2010).

Our results indicate that property and sales taxes appear insufficient to spur PV market deployment from 1997 to 2009. State-level property tax incentives tend to be quite limited in scope, given that the property tax is most often assessed at the local-level and we did not track local property tax incentives. In addition, property tax incentives are structured such that the market value of solar installations is not included in the total assessment of property values. In the absence of these incentives, installing solar technologies may incur a penalty through increased property values. Given that avoiding a penalty may be less of an incentive compared to a direct incentive, such as an income tax credit or cash rebate, one would expect that property tax incentives would be individually less effective than income tax or cash incentives. Similarly, sales taxes are typically less than 10% of the sales price for the technology (before installation), whereas income tax and cash incentives are typically more than 10% and sometimes as high as 30% of the total cost including installation. Given the relative size of the incentives, one would expect the sales taxes to be less significant individually compared to income tax and cash incentives.

In sum, our results suggest that PV markets have been responsive to the simplicity of the financial incentive offered and the size of the resultant incentive. Cash incentives tend to be simple and large in value, and were found to exhibit systematic

Table 2
Estimation results.

Variable	Model (1): PV capacity-installed logged, 1997–2009 (kW)	Model (2): PV capacity-installed logged, 1997–2009 (kW)
ITI/YRITI	−0.25 (0.26)	0.07 (0.05)
CI/YRCI	1.29*** (0.28)	0.21*** (0.06)
PTI/YRPTI	−0.41 (0.43)	0.04 (0.04)
STI/YRSTI	0.50 (0.33)	0.03 (0.05)
RPS/YRRPS	0.71*** (0.27)	0.16*** (0.06)
CARVE/YRCARVE	2.19*** (0.40)	0.20 (0.13)
METER/YRMETER	0.30 (0.25)	0.06 (0.05)
ESCCD	2.03** (0.95)	2.27** (0.97)
PCGDP	−3.38* (1.84)	−5.42*** (1.90)
TPOPP	−0.68 (2.69)	2.03 (3.12)
State dummy variables	Yes	Yes
Time dummy variables	Yes	Yes
R-squared (with state and time effects)	0.64	0.64
R-squared (without state or time effects)	0.34	0.31
Observations	650	650
Wald-test	$\chi^2(71)=3131.95$	$\chi^2(71)=3168.46$

Note: Robust standard errors in parentheses.

* $p < .10$.

** $p < .05$.

*** $p < .01$.

effects on PV deployment; whereas tax incentives (income, property, and sales) were found not to exhibit such systematic effects perhaps because they tend to be smaller in value and suffer from implementation complexity.

In addition to the financial incentives, we found that state *RPS* policies were important in promoting solar deployment, and were most effective when coupled with specific carve-out provisions for solar or distributed energy. This result confirms the findings of previous case-study research (Wiser and Barbose, 2008) and econometric studies of other renewable technologies (Yin and Powers, 2010).

We found that states with *RPS* policies installed an average of 95% more PV capacity than states without *RPS* policies, while states with specific solar set-asides installed over 722% more PV capacity, on average, than states without solar carve-outs. Our results highlight the apparent effectiveness of mandatory solar power requirements relative to financial incentives in promoting market deployment. Further, the results suggest that, in the absence of a solar carve-out provision, states may favor other, lower-cost renewable energy sources to meet their *RPS* requirements such as wind turbines. If the states want to promote a diversified portfolio of renewable sources, including solar, specific set-asides for solar or distributed technologies appear crucial.

In addition, our results suggest that PV installations increased with every additional year that a state offered a cash incentive, or had an *RPS* or solar carve-out provision. The coefficients in Model 2 are consistent but smaller than in Model 1. The coefficients in Model 1 illustrate the average effect of having an incentive over the entire study period while the coefficients in Model 2 illustrates the incremental effect of having a policy for an additional year. The Model 2 results suggest that these policies became more effective over time, perhaps as program implementers gain experience and consumers and installers became more familiar with available incentives. The duration variables may also capture the stringency of the underlying policy — as in case of *RPS*, where the *RPS* requirement typically increased between the effective date and future target date, driving more deployment in later years. Taken together, the results suggest that states with cash incentives or with *RPS* requirements had stronger PV deployment than states that did not have these policies, and that PV deployment increased more rapidly over time than in states without these policies.

Our results revealed that consumers were more likely to install PV technology in states with higher electricity prices, such as Hawaii and the New England states. The return on investment for solar technologies improves when electricity prices are higher, resulting in larger future electricity cost savings. The results suggest that PV capacity installed increased by approximately 2% for every 1% increase in electricity prices, on average, during the 1997–2009 period. Thus, our results hint that a doubling of electricity prices would have had a stronger effect on PV deployment during our study period than adopting an *RPS*, although less of an effect than adopting a cash incentive.

The results indicate that consumers in wealthier states were somewhat less likely to adopt PV technologies during the study period, after controlling for other variables. As mentioned earlier, little systematic information exists on the wealth of actual PV customers, which would be a better measure of willingness or ability to pay for solar technology. In addition, much of the demand for PV lies with commercial or industrial customers that may be less income-constrained than individual consumers.

The model does not reveal significant population scaling effects (*TPOPP*) after controlling for other variables. The state and time dummy variables captured a large amount of variation across states and over time, with states typically exhibiting consistently higher PV deployment than Alaska (the reference

state) and with annual PV deployment consistently growing over time as compared with deployment in 1997. Last, the regression results are robust to different specifications and the omission of the time dummies.

2. Discussion

Existing research on the effect of multiple state-level financial incentives on the market deployment of solar technology has not kept up with the increasing interest in solar from consumers, utilities, and legislators. In this paper, we used 13 years of state-level data to examine the effect of four types of financial incentives offered by states.

Overall, we found that states offering cash incentives had significantly stronger market deployment of PV than states not offering cash incentives, while states offering tax incentives (income, property, and sales) did not have systematically stronger deployment than other states. Further, we observe that cash incentives and *RPS* increased solar PV deployment on average across the full study period but also that each additional year of these policies resulted in incrementally higher deployment.

Several caveats of our analysis must be noted. In this paper, states with multiple incentives of the same type are treated the same as states with just one. In addition, the scale, scope, and strength of each incentive are not taken into account. Further research needs to investigate the total value of incentives within a state for example installations, such as for particular end-use sectors that will simplify the analysis (e.g., Wiser et al., 2011). Further analysis could also examine the suite of public, private, and non-profit financial incentives available in selected locations currently and moving forward, such as in Oregon and North Carolina where non-profit incentives appeared to have spurred some PV deployment. Future analysis could also map the availability of current local or utility-specific incentives across a state, with a goal to illustrate the availability of incentives at a sub-state level.

Issues with endogeneity may arise from the presence of special interest groups that lobby for policies favorable to solar technology. In this case, the presence of an incentive may not by itself be the cause of market deployment, and any apparent correlations with the policy tools would be inappropriately attributed to the incentive. Such a problem is present in all studies of policy effectiveness (Lazzari, 1983). We acknowledge this potential problem and caution that our results illustrate contemporaneous correlation of policy with market deployment but do not guarantee causation. Nevertheless, we are confident that states with cash incentives supporting solar technology and renewable energy mandates have experienced stronger market deployment of PV technologies than states that do not offer these policies. If states want to encourage solar technology, they must consider how the full complement of policy tools offered in their state affects the likelihood of solar technology deployment.

Our analysis ends with the year 2009. Beginning in the last quarter of 2008, the global financial crisis likely discouraged some solar installations by price-conscious consumers. In addition, the fiscal capacity of many states has been severely compromised in recent years. Federal dollars from the American Recovery and Reinvestment Act of 2009 were used to support some existing or new financial incentives for alternative energy in states not traditionally offering incentives. As the federal recovery money runs out and as the difficult fiscal situation continues, some states have reduced or discontinued their incentives for solar energy. This volatile policy and funding environment makes rigorous econometric assessments of state-level financial incentives challenging. We included 2009 as the latest year of complete

information but acknowledge that the situation may have been somewhat different in 2009 than earlier years.

Finally, future analysis will need to closely consider the myriad ways in which financial incentives have changed in design and implementation in recent years. For instance, states are now developing private markets for sale of renewable energy credits and encouraging long-term power purchase agreements with utilities as primary tools for encouraging solar PV deployment. These tools shift the financing burden at least somewhat from states to utilities and private entities. States are also experimenting with more complex financing arrangements, including third-party payer arrangements, which facilitate deployment by shifting the upfront costs from individual private customers to investors willing to share the burden. The complexities of these new policy tools are not reflected in the present analysis but highlight the need for continued evaluation and analysis of state-level policy innovation moving forward.

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